PRACTICAL COURSE WORKBOOK

AI in Healthcare & Diagnostics

Work with digital-health information responsibly

LEVEL	Intermediate
GUIDED TIME	5 hours across 12 lessons
PROJECT	a safe, fictional health-data workflow and review
SKILLS	Medical AI, Imaging, Clinical NLP, FDA AI, Predictive models, Critical evaluation
EDITION	Structured draft - version 0.9

WHAT YOU WILL BE ABLE TO DO

- Use Medical AI appropriately in a realistic, bounded task.
- Use Imaging appropriately in a realistic, bounded task.
- Use Clinical NLP appropriately in a realistic, bounded task.
- Use FDA AI appropriately in a realistic, bounded task.

WHO THIS IS FOR

Learners exploring health technology without replacing clinical expertise.

BEFORE YOU BEGIN

- Basic data literacy

Use this workbook with the online lessons. It is designed for A4 printing, handwritten evidence notes, and offline revision. It does not provide certification.

WORKBOOK GUIDE

A clear route through the course

AI in Healthcare & Diagnostics is a structured, practical course covering Medical AI, Imaging, Clinical NLP, FDA AI, Predictive models. It emphasizes explainable methods, checked work and a final outcome that can be reviewed.

MODULE	LESSONS AND FOCUS
01 Medical AI: Health context	1. Care pathways with Medical AI 2. Health data with Imaging 3. Stakeholders with Clinical NLP
02 Imaging: Digital systems	1. Standards with FDA AI 2. Interoperability with Predictive models 3. Workflows with Medical AI
03 Clinical NLP: Quality and safety	1. Validation with Imaging 2. Privacy with Clinical NLP 3. Human oversight with FDA AI
04 FDA AI: Applied evaluation	1. Implementation with Predictive models 2. Monitoring with Medical AI 3. Equity with Imaging

HOW TO USE EACH LESSON PAGE

1. Read the objective and identify the decision it supports.
2. Attempt the retrieval prompt before checking the online explanation.
3. Reproduce the worked example with the stated input.
4. Test one normal case and one boundary or failure case.
5. Record the result, limitation and next action in the evidence log.

PRINT AND FILE

Print at 100% scale on A4 paper. Use double-sided printing with long-edge binding. Keep completed pages with the project files named in the evidence log.

MODULE 01

Medical AI: Health context

Apply Medical AI through care pathways, health data, stakeholders.

LESSON 01 / 18 MIN

Care pathways with Medical AI

Learning objectives

- Explain care pathways in the context of AI in Healthcare & Diagnostics.
- Apply Medical AI to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

care pathways, Medical AI, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a care pathways result produced with Medical AI?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 02 / 22 MIN

Health data with Imaging

Learning objectives

- Explain health data in the context of AI in Healthcare & Diagnostics.
- Apply Imaging to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

health data, Imaging, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a health data result produced with Imaging?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 03 / 26 MIN

Stakeholders with Clinical NLP

Learning objectives

- Explain stakeholders in the context of AI in Healthcare & Diagnostics.
- Apply Clinical NLP to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

stakeholders, Clinical NLP, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a stakeholders result produced with Clinical NLP?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 02

Imaging: Digital systems

Apply Imaging through standards, interoperability, workflows.

LESSON 04 / 30 MIN

Standards with FDA AI

Learning objectives

- Explain standards in the context of AI in Healthcare & Diagnostics.
- Apply FDA AI to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

standards, FDA AI, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a standards result produced with FDA AI?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 05 / 18 MIN

Interoperability with Predictive models

Learning objectives

- Explain interoperability in the context of AI in Healthcare & Diagnostics.
- Apply Predictive models to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

interoperability, Predictive models, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a interoperability result produced with Predictive models?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 06 / 22 MIN

Workflows with Medical AI

Learning objectives

- Explain workflows in the context of AI in Healthcare & Diagnostics.
- Apply Medical AI to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

workflows, Medical AI, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a workflows result produced with Medical AI?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 03

Clinical NLP: Quality and safety

Apply Clinical NLP through validation, privacy, human oversight.

LESSON 07 / 26 MIN

Validation with Imaging

Learning objectives

- Explain validation in the context of AI in Healthcare & Diagnostics.
- Apply Imaging to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

validation, Imaging, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a validation result produced with Imaging?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 08 / 30 MIN

Privacy with Clinical NLP

Learning objectives

- Explain privacy in the context of AI in Healthcare & Diagnostics.
- Apply Clinical NLP to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

privacy, Clinical NLP, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a privacy result produced with Clinical NLP?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 09 / 18 MIN

Human oversight with FDA AI

Learning objectives

- Explain human oversight in the context of AI in Healthcare & Diagnostics.
- Apply FDA AI to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

human oversight, FDA AI, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a human oversight result produced with FDA AI?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

MODULE 04

FDA AI: Applied evaluation

Apply FDA AI through implementation, monitoring, equity.

LESSON 10 / 22 MIN

Implementation with Predictive models

Learning objectives

- Explain implementation in the context of AI in Healthcare & Diagnostics.
- Apply Predictive models to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

implementation, Predictive models, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a implementation result produced with Predictive models?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 11 / 26 MIN

Monitoring with Medical AI

Learning objectives

- Explain monitoring in the context of AI in Healthcare & Diagnostics.
- Apply Medical AI to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

monitoring, Medical AI, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a monitoring result produced with Medical AI?

Evidence to keep

[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

LESSON 12 / 30 MIN

Equity with Imaging

Learning objectives

- Explain equity in the context of AI in Healthcare & Diagnostics.
- Apply Imaging to a bounded practical task.
- Evaluate the result using explicit quality criteria.

Key terms

equity, Imaging, healthcare

Retrieval prompt

In AI in Healthcare & Diagnostics, which evidence best supports a equity result produced with Imaging?

Evidence to keep	[ ] Original input [ ] Observed result [ ] Boundary or failure result [ ] Limitation and next action

MODULE REVIEW

Turn the module into usable evidence

1. RETRIEVE

Without checking the lessons, explain the module's three most important ideas in your own words.

2. APPLY

Describe the example or file you produced, the input used, and the result you observed.

3. REVIEW

Record one uncertainty, one source to recheck, and the next deliberate practice action.

CAPSTONE PROJECT

Produce a reviewable AI in Healthcare & Diagnostics project using Medical AI, Imaging, Clinical NLP.

DELIVERABLE	A working AI in Healthcare & Diagnostics artefact plus an evidence-based self-review.
TOOLS	A suitable Medical AI environment, A plain-text decision log, Test data or realistic sample material

Production plan

1. Define the intended user, outcome and constraints.
2. Create the smallest complete result using Medical AI.
3. Test one normal case, one boundary case and one failure response.
4. Revise the work from the evidence and preserve before-and-after results.
5. Prepare a concise handover containing method, limitations and next step.

Definition of done

- The output matches the stated outcome.
- Inputs and decisions are reproducible.
- Boundary and failure evidence is included.
- Limitations and responsibility considerations are explicit.

RESPONSIBLE PRACTICE

This material is educational and does not provide medical advice or replace qualified clinical judgement.

PROJECT EVIDENCE

Document decisions another person can review

1. PURPOSE AND USER
State the intended user, need, expected outcome and constraints.

2. INPUTS AND ASSUMPTIONS
List source files, data, versions, permissions and assumptions.

PROJECT EVIDENCE

Tests, limitations and revision

3. NORMAL CASE
Expected result, observed result and evidence location.

4. BOUNDARY OR FAILURE CASE
What was difficult, what happened and why it matters.

5. REVISION AND LIMITATION
What changed, what improved and what remains unproven.

REVIEW AND SOURCES

Make the work credible and repeatable

Final self-review

- Can another learner repeat the method?
- Did I test a difficult case?
- Did I avoid unsupported claims?
- Is the next action proportionate to the remaining risk?

FINAL DECISION
Is the project ready to share? State the evidence, remaining risk, owner and next review date.

Authoritative references

Global strategy on digital health 2020-2027

World Health Organization - reviewed 2026-08-21

https://www.who.int/publications/i/item/9789240116870

FHIR specification

HL7 International - reviewed 2026-08-21

https://hl7.org/fhir/

SOURCE NOTE

Product versions, laws and professional standards can change. Recheck the linked primary source before applying version-sensitive information.

NEXT IMPROVEMENT

Choose one weakness found during review and improve it before extending the Medical AI scope.